Automated Conference Management Application

Software Requirements Specification

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Group - B

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to present a detailed description of the **Automated Conference Management Application**, a web-based application developed to support the planning, management and execution of academic conferences. This document defines the system's major features and workflows such as conference creation, user registration, paper submission and tracking, participant registration, session scheduling, committee and task management, email notifications and report generation.

This document also specifies the system interfaces, operating constraints, assumptions, dependencies, design constraints and limitations (inverse requirements) under which the system must function, including the use of external services such as email servers, payment gateways and third-party review/plagiarism platforms (e.g., EasyChair).

The document is prepared as part of a class project under the Software Engineering course and if successfully implemented, the system may be deployed for conference management activities at IIT Indore.

1.2 Scope

The software product to be developed is the **Automated Conference Management Application**. It is a web-based application intended to support the end-to-end organization and management of academic conferences at an institutional level. The system provides a centralized platform to reduce manual workload using automation, improve coordination between organizing teams and ensure proper tracking of conference activities.

1.2.1 Software Products / Modules Included

The Conference Organization System includes the following major modules:

- Conference Management Module
- User Registration Module
- Paper Submission & Status Tracking Module
- Review & Decision Support Module (Integration-based)
- Participant Registration Module
- Session Scheduling & Conflict Detection Module
- Committee & Task Management Module
- Logistics Management Module (venue, seating, resources)
- Food and Accommodation Management Module

- Email Notification and Reminder Module
- Reporting Module

1.2.2 What the System Will Do

The Conference Organization System will:

- Allow organizers to create and manage conference details such as dates, venue/location, the-me and status.
- Support role-based user access for Organizers, Chairs, Reviewers, Authors and Participants.
- Allow authors to submit papers (PDF upload), track submission status and upload revised versions.
- Support reviewer assignment and review tracking while integrating with external review/plagiarism platforms where required.
- Allow participant registration and classification into Regular Participants and Presenters based on chair-approved decisions.
- Enable scheduling of conference sessions and detect scheduling conflicts.
- Manage conference logistics including venue allocation, seating arrangements and resource planning.
- Manage food and accommodation requirements for participants.
  Send event-based and scheduled notifications such as reminders, invitations and follow-ups through email services.
- Generate administrative reports such as submission statistics, acceptance rates and logistics readiness.

1.2.3 What the System Will NOT Do

The Conference Organization System will not:

- Make final academic decisions automatically; paper acceptance/rejection requires Chair approval.
- Perform direct payment transactions; payment will be handled by external gateways and the system will only track payment confirmation.
- Replace external platforms such as EasyChair or plagiarism tools; it will only integrate with and use outputs from them.
- Publish certificates, websites, or automated communications without explicit organizer approval.
- Expose confidential reviewer information, plagiarism reports, or internal committee discussions to unauthorized users.

1.2.4 Application, Objectives and Benefits

The application of this system is to improve conference organization efficiency by

automating repetitive tasks and maintaining a structured workflow for all stakeholders. The main objectives and benefits include:

- Faster operations and reduced manual effort by organizing all conference activities in one platform.
- Improved traceability of papers, decisions, committee tasks and logistics through uniquely tracked records.
- Reliable communication through automated email triggers and scheduled reminders.
- Better planning and coordination using scheduling conflict detection and committee task tracking.
- Secure access control through authentication and role-based authorization ensuring confidentiality of sensitive data.

1.3 References

- https://easychair.org/
- https://conferences.ieee.org/conferences_events/

1.4 Overview

This Software Requirements Specification (SRS) document provides a complete description of the requirements for the Conference Organization Automation System. It includes the system goals, major features, user roles, constraints and detailed requirements needed for designing, implementing and testing the application.

This SRS is organized as follows:

- **Section 1: Introduction** – explains the purpose, scope and organization of this SRS document.
- **Section 2: General Description** – gives an overall view of the system including product perspective, product functions, user characteristics, constraints and assumptions.
- **Section 3: Specific Requirements** – defines detailed functional and non-functional requirements, system classes/objects, design constraints, inverse requirements, database requirements and additional requirements such as reporting and automation rules.

2. General Description

This section describes the general factors that affect the **Automated Conference Management App** and its requirements. This section does not state specific requirements but provides an overall understanding of the system, its environment, users, constraints and assumptions.

2.1 Product Perspective

The **Automated Conference Management App** is a modular, web-based software application designed to support the end-to-end organization and management of academic conferences. The system functions as a centralized platform that integrates multiple conference-related workflows, including paper submission, internal committee coordination, scheduling, logistics, outreach and participant management.

The system operates primarily as a standalone application but interacts with several external services such as email platforms, plagiarism detection systems (e.g., EasyChair), calendar services, website hosting platforms and online payment gateways. It supports offline conferences accounting for time-zone differences and distributed participation

2.2 Product Functions

The major functions of the Automated Conference Management App include:

- Managing conference metadata, important dates, schedules and sessions
- Automated generation of the conference website
- User registration, authentication and role-based access control
- Paper submission, plagiarism screening integration, review tracking and decision recording
- Participant registration and classification (Presenter / Regular Participant)
- Internal committee management (Review, Outreach, Logistics, Food, Registration)
- Task assignment, acceptance tracking and progress monitoring for committees
- Calendar management with conflict detection for sessions, speakers and venues
- Logistics management including venue allocation
- Food and catering management with participant feedback
- Automated outreach to speakers and authors with follow-ups
- Tracking registration payments via external payment gateways
- Sending system notifications, reminders and approved communications
- Automated certificate generation and feedback collection
- Generating administrative, operational and analytical reports

2.3 User Characteristics

The users of the Automated Conference Management App include:

- **Organizers / Admins:** Manage the overall conference, configure workflows and oversee automation
- **Conference Chairs:** Make final decisions on paper acceptance, reviewer assignments and presenter approvals
- **Reviewers:** Evaluate submitted papers and provide structured feedback
- **Authors:** Submit papers and track review outcomes
- **Participants:** Attend the conference as regular participants or presenters

Users are expected to have basic familiarity with computers and web-based applications. No advanced technical expertise is required. The system is designed with intuitive dashboards tailored to each user role.

2.4 General Constraints

The system is subject to the following constraints:

- The system must operate on standard modern web browsers
- Continuous internet connectivity is required for core functionalities
- External services (email APIs, plagiarism tools, calendar services) must be available
- Development must adhere to institutional and academic guidelines
- The project is constrained by academic timelines and available development resources
- The system is optimized for college to mid-scale academic conferences

2.5 Assumptions and Dependencies

- The system assumes that users have access to internet-enabled devices such as laptops or smartphones. It depends on the availability and reliability of external services including email servers, plagiarism checking platforms, calendar APIs, website hosting services and payment gateways.
- It is assumed that final academic decisions (paper acceptance, presenter invitations) will always be made by human conference chairs. Any changes in these assumptions or dependencies may require corresponding updates to the system requirement

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

- **Organizer Dashboard:** A central interface for conference setup (date, topic, department), managing logistics (accommodation, food) and committee management portal.
- **Committee Management Portal:** An interface to create job roles for team leads and monitor their acceptance status. Create tasks for the team and monitor progress.
- **Speaker Invitation Suite:** A text-editor interface integrated with an LLM to generate and track customized invitation emails.
- **Paper allotment Portal:** An interface accessed by the organizer to allot research papers for review to the reviewers.
- **Review Portal:** An interface for reviewers to access papers without author metadata and give score and comments.
- **Participant Submission & Registration:** An interface for paper uploads with real-time format validation and a conditional registration gateway for accepted authors.

3.1.2 Software Interfaces

- **LLM API (e.g., OpenAI/Gemini):** The system shall interface with an LLM via API to generate customized emails for speakers, classify feedback and reply mails.
- **Email Service Provider (SMTP/API):** The system shall interface with a mail server to send automated updates, reminders and certificates to users.
- **PDF Generation Library:** The system shall utilize a server-side library to dynamically generate PDF certificates for participants upon completion of the feedback form.

3.1.3 Communications Interfaces

- **Network Protocol:** The system shall communicate over HTTPS to ensure secure data transfer between the laptop browser and the server.
- **Email Protocols:** The system shall use standard mail protocols (SMTP) for all outgoing reminders and confirmation updates.

3.2 Functional Requirements

3.2.1 Automated Paper Review Management

- **Introduction:** This feature manages the lifecycle of a research paper from submission to decision.
- **Inputs:** Research paper files (PDF), department keywords.
- **Processing:**
    1. Validate file size and formatting (indentation, text format).
    2. Allot papers to reviewers based on department matching.
    3. Aggregate multiple reviewer scores into a final decision metric.
- **Outputs:** Accept/Reject notifications sent to participants and anonymized feedback reports.

- **Error Handling:** If a paper fails format validation, the system shall provide an immediate rejection notice with specific formatting errors listed.

3.2.2 LLM-Powered Communication & Classification

- **Introduction**: This feature automates speaker invitation, follow-ups and reply analysis using AI.
- **Inputs**:
  Conference theme and speaker requirements, speaker profiles and contact information.
- **Processing**:
  1. Generate personalized invitation emails.
  2. Schedule email dispatch and follow-ups.
  3. Analyze incoming responses using LLM-based classification.
- **Outputs**:
  Sent invitation and follow-up emails, categorized responses (Accepted / Rejected / No Response).
- **Error Handling**:
  If email delivery fails, the system shall retry sending and log the failure for organizer review.

3.2.3 Internal Committee Management

- **Introduction**: This feature manages internal committees involved in conference organization.
- **Inputs**: Committee type and job details, selected committee members.
- **Processing**:
  1. Circulate committee invitations.
  2. Track accept/reject responses.
  3. Assign tasks and monitor progress.
- **Outputs**: Committee formation status and task and tracking dashboards.

3.3 Classes / Objects

3.3.1 User

3.3.1.1 Attributes

- user_id : Integer – Unique identifier for the user
- name : String – Full name of the user
- email : String – Registered email address
- password : String – Encrypted password
- affiliation : String – Institution or organization

3.3.1.2 Functions

- register() – Registers a new user in the system
- login() – Authenticates user credentials
- updateProfile() – Updates user details

- logout() – Ends user session

3.3.2 Conference

3.3.2.1 Attributes

- conference_id : Integer – Unique identifier for the conference
- conference_name : String – Name of the conference
- theme : String – Main theme or topic of the conference
- start_date : Date – Conference start date
- end_date : Date – Conference end date
- location : String – Venue or online platform
- conference_type : Enum – Online / Offline / Hybrid
- organizer_contact : String – Organizer email or contact number
- status : Enum – Upcoming / Ongoing / Completed

3.3.2.2 Functions

- createConference() – Creates a new conference entry
- updateConference() – Modifies conference details
- getConferenceDetails() – Retrieves conference information
- setConferenceStatus() – Updates current conference status

3.3.3 Paper

3.3.3.1 Attributes

- paper_id : Integer – Unique identifier of the paper
- title : String – Title of the paper
- file_path : String – Uploaded paper file (PDF)
- status : Enum – Submitted / Screening / Under Review / Accepted / Rejected
- author_id : Integer – Reference to submitting user

3.3.3.2 Functions

- submitPaper() – Submits a new paper
- uploadFile() – Uploads paper PDF
- updatePaper() – Uploads revised version
- getStatus() – Returns current paper status

3.3.4 Committee

3.3.4.1 Attributes

- committee_id : Integer – Unique identifier for the committee
- conference_id : Integer – Reference to the associated conference
- committee_type : Enum – Review / Logistics / Outreach / Food / Registration
- description : String – Committee responsibility summary
- created_date : Date – Committee creation date
- status : Enum – Active / Inactive

3.3.4.2 Functions

- createCommittee() – Creates a committee for a conference
- updateCommittee() – Modifies committee details
- assignMembers() – Assigns members to the committee
- viewCommitteeDetails() – Displays committee information

3.3.5 Committee Task

3.3.5.1 Attributes

- task_id : Integer – Unique identifier for the committee task
- committee_id : Integer – Reference to parent committee
- conference_id : Integer – Reference to associated conference
- task_title : String – Name of the task
- task_description : Text – Detailed task information
- assigned_member_ids : List – Committee members assigned
- deadline : Date – Task completion deadline
- status : Enum – Pending / In Progress / Completed

3.3.5.2 Functions

- createTask() – Creates a task under a committee
- assignTask() – Assigns task to committee members
- updateTaskStatus() – Updates task progress
- viewTaskStatus() – Displays current task status

3.3.6 Session

3.3.6.1 Attributes

- session_id : Integer – Unique session identifier
- session_title : String – Name of the session
- session_type : Enum – Keynote / Workshop / Panel
- date_time : DateTime – Scheduled time
- venue : String – Assigned venue
- speaker_id : Integer – Speaker reference

3.3.6.2 Functions

- createSession() – Creates a new session
- updateSession() – Modifies session details
- checkConflicts() – Detects schedule conflicts

3.3.7 Email Generator

3.3.7.1 Attributes

- generator_id : Integer – Unique identifier for the email generator
- conference_id : Integer – Reference to associated conference
- template_id : Integer – Reference to email template
- recipient_list : List – List of recipient email addresses
- email_subject : String – Generated email subject
- email_body : Text – Generated email content
- email_type : Enum – Notification / Reminder / Invitation / Follow-Up
- send_status : Enum – Pending / Sent / Failed

3.3.7.2 Functions

- generateEmail() – Generates email content using predefined templates
- sendEmail() – Sends generated email to recipients
- sendBulkEmail() – Sends the same email to multiple recipients
- scheduleEmail() – Schedules email for later delivery
- resendEmail() – Retries sending failed emails

3.4 Non-Functional Requirements

3.4.1 Performance

- Paper upload operations (PDF ≤ 20 MB) shall complete within 5 seconds for 90% of submissions.
- Email generation and dispatch requests shall be triggered within 3 seconds of the event occurrence.
- Dashboard pages (author, reviewer, organizer) shall load within 3 seconds for 95% of users.

3.4.2 Reliability

- No submitted paper, review, or committee task data shall be lost during system failures.
- In case of email delivery failure, the system shall automatically retry.

3.4.3 Availability

- Core services (login, paper submission, status tracking) shall be accessible 24×7 during the conference lifecycle.
- Automated background services (email reminders, follow-ups) shall operate continuously without user intervention.

3.4.4 Security

- All users shall be authenticated using secure login with encrypted credentials (JWT/session-based authentication).
- Role-based access control shall be enforced for Authors, Reviewers, Organizers and Committee Members.
- The system shall prevent unauthorized access to reviews, decisions and committee data.

3.4.5 Maintainability

- The system shall follow a modular architecture, allowing individual modules (submission, email, logistics, committee) to be updated independently.
- Configuration changes (email templates, deadlines, conference details) shall not require source code modification.

3.4.6 Portability

- The system shall be accessible on modern web browsers (Chrome, Firefox, Edge) without requiring additional plugins.
- No client-side installation shall be required; the system shall function entirely via a web interface.

3.5 Inverse Requirements

The system shall **not**:

- Make final academic decisions autonomously; all paper acceptance, rejection, or revision decisions shall require explicit approval by Conference Chairs.
- Issue presenter invitations unless a paper has completed the submission, plagiarism screening, review and Accepted decision workflow.
- Exclude authors from conference participation solely due to paper rejection; rejected authors shall be permitted to register as Regular Participants.
- Expose confidential artifacts such as reviewer identities, reviewer comments, plagiarism reports, or internal committee discussions to unauthorized users or the public.
- Replace external plagiarism or review platforms (e.g., EasyChair); the system shall only integrate with and consume outputs from such platforms.
- Publish conference websites, abstract booklets, certificates, or automated communications without explicit organizer approval.

- Perform direct financial transactions; payment processing shall be handled by external gateways, with this system tracking only registration status and payment confirmation.

3.6 Design Constraints

The following constraints apply to the design and implementation of the system:

- **Architecture Constraint**: The system shall follow a web-based, modular architecture with a React-based frontend, a Node.js backend and a relational database (e.g., MySQL).
- **Integration Constraint**: Plagiarism detection and, where applicable, review workflows shall depend on EasyChair and its supported tools (e.g., Turnitin/iThenticate).
- **Automation Constraint**: Workflow automation (website generation, email reminders, certificate generation, feedback collection) shall be implemented using orchestration tools such as n8n and scheduled background jobs.
- **Security Constraint**: Authentication and authorization shall be implemented using secure mechanisms (JWT or session-based) with strict role-based access control.
- **Operational Constraint**: The system is optimized for small to mid-scale academic conferences (college or institutional level) and is not required to support enterprise-scale concurrency.
- **Policy Constraint**: System behavior must align with institutional IT, academic integrity and data privacy policies applicable to academic conferences.

3.7 Logical Database Requirements

The system shall use a centralized relational database to support all functional modules. Logical requirements include:

- **Core Entities**:
    - Users (roles: Organizer, Chair, Reviewer, Author, Participant)
    - Papers, Paper Versions and Submission Metadata
    - Reviews, Reviewer Assignments and Decision Records
    - Participant Categories (Regular Participant, Presenter)
    - Sessions, Calendars, Venues, Logistics and Catering Records
    - Committees, Tasks, Budgets, Expenses and Audit Logs
    - Outreach Records, Email Logs, Certificates and Feedback
- **Integrity Rules**:
    - Each paper shall have a unique identifier and complete lifecycle traceability.
    - Presenter status shall be assigned only when a paper decision is marked Accepted by a Chair.
    - Referential integrity shall be enforced across users, submissions, reviews, sessions and committees.
- **Retention Requirements**:
    - Submission data, reviews and decisions shall be retained for the full conference lifecycle.
    - Financial and committee audit logs shall be preserved for reporting and accountability.
- **Access Control**:

    - Confidential data (reviews, plagiarism reports, internal notes) shall be logically separated from public data (schedule, accepted papers).

3.8 Other Requirements

- **Participant Classification Requirement**:
    - The system shall support two participant categories:
        1. **Regular Participants** – attendees without presentation privileges.
        2. **Presenters** – authors whose papers are accepted by the Conference Chairs.
    - Authors whose papers are rejected shall automatically be offered registration as Regular Participants.
- **Chair Authority Requirement**:
    - Conference Chairs shall retain final authority over paper decisions, reviewer assignment overrides and presenter invitations.
- **Human-in-the-Loop Automation Requirement**:
    - All AI-assisted or automated actions (email drafting, summaries, reminders) shall be reviewable, editable and overridable by authorized organizers.
- **Reporting Requirement**:
    - The system shall generate administrative reports covering submission statistics, acceptance rates, participant categories, logistics readiness and feedback summaries.
- **Extensibility Requirement**:
    - The system shall support future extensions (e.g., hybrid conference support, advanced analytics) without significant architectural changes.
- **Traceability Note**:
    - These requirements trace directly to functional modules described in Sections 3.2 and 3.3, including paper submission, committee management, outreach automation, scheduling, logistics and reporting